

Development of variational quantum deep neural networks for image recognition

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ABSTRACT

Parametrized quantum circuits are widely used for supervised learning tasks such as image classification in the noisy intermediate scale quantum era. However, normally, it can only handle low-dimensional data. This study presented a variational quantum deep neural network (VQDNN) model for various scale image recognition tasks. Three classifiers were designed to verify the classification performance of the proposed VQDNN model. In the first classifier, to accommodate the limitations of qubits in both simulation hardware and real quantum hardware – we adopted hybrid principal component analysis – VQDNN architecture. Moreover, the amplitude encoding scheme and the rotation angle coding scheme were employed in the subsequent two classifiers to handle large-size images. Finally, we used the classical neural network and VQDNN model to conduct a comparative experiment of the ten-label classification learning task on the same dataset. The quantum numerical experiment was implemented on two benchmark datasets: the MNIST and UCI databases of handwritten digits. The simulation results showed that the proposed VQDNN classified the two datasets with an accuracy of 100% for the two-class classification task, while the UCI dataset has an accuracy of 90.87% for the ten-label classification task. The proposed VQDNN achieved better classification accuracy than the original classical neural network even under a limited number of qubits available in current hardware, indicating the promising application potential of VQDNN in image recognition.

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1. Introduction

The field of quantum computing originated from the recognition that classical simulation of the quantum system is a challenging task [14]. Studying an alternative to the classical algorithm based on the concept of quantum superposition and entanglement has greatly. Additionally, the novel devices, called Noisy Intermediate-Scale Quantum (NISQ) computers, have increasingly attracted much attention [37]. It was shown that the state-of-the-art NISQ outperforms the classical supercomputers in contrived mathematical tasks [1,48].

Variational quantum algorithms (VQAs) have emerged as a promising approach to obtain a quantum advantage on NISQ [10]. As a natural quantum analog of neural networks, VQAs adopt an optimization-based method. The classical optimization toolbox

is used to run the parameterized quantum circuits on the quantum computer, and then a classical optimizer is used to optimize the parameters. In such a way, the quantum circuit depth can remain shallow, mitigating noise, unlike quantum algorithms developed for the fault-tolerant era. Due to this advantage, VQAs have been used for various quantum computer applications, especially universal quantum computing [8], dynamical quantum simulation [13,33,3], optimization [38,30,34], mathematical applications [44,29,19], and quantum machine learning.

Quantum machine learning aims to learn patterns for predicting unseen data accurately. An excellent review on variational or parametrized circuits in quantum machine learning can be found in Ref. [6].

Expressiveness is the central topic of classical machine learning, and it has aroused great interest in quantum machine learning. Although parametrized quantum circuits (PQC) provides strong evidence of quantum advantage, there are still two important questions that have not been explored: (1) How can a high expressive PQC be designed? (2) Does PQC have any quantum advantages

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that can be used to solve practical problems? The comparison of expressive power between PQC and classical neural networks is desirable and may be beneficial to the fields of physics and machine learning. There have been some publications on this aspect of research, such as the structure of variational quantum neural networks [5,11,35], and quantum neural networks as classifiers [26,45].

A proof-of-principle experiment has proved that PQCs with a simple structure already outperform any classical neural network for learning tasks in Ref. [12]. However, the research mainly focuses on generative learning tasks. Schuld et al. [42] investigated how the strategy with which data are encoded into the model influences the expressive power of parametrized quantum circuits as function approximators. The results show that by repeating simple data-encoding gates multiple times, quantum models can access increasingly rich frequency spectra. In Ref. [16], a parameterized tree-like quantum classifier was proposed. The classifier can achieve decent performance for the Iris dataset and MNIST with proper strategies. However, the research used dimension reduction methods such as principal component analysis (PCA) to reduce 784 dimensional data to an eight-dimensional vector, which will inevitably cause some information loss. Schuld et al. [39] proposed a parameterized circuit model with a circuit composed of variational single-qubit gates and two-qubit controlled gates. This work provides several fresh angles to examine variational quantum classifiers. In addition to the above structures and protocols of variational quantum classifiers, there are many other works from different perspectives [9,25,47,22].

After careful analysis, it was determined that the previous research still needs more in-depth evaluation in at least the following four aspects:

- To process real datasets, most of the literature uses principal component analysis (PCA) to preprocess the images and project real data to low dimensions, which reduces the accuracy of the algorithm.
- Most documents only use one coding method and do not compare the impact of different coding techniques on the accuracy of the classification model.
- Most of the literature only performs a kind of binary classification, and there is no systematic comparison of model classification effects under different learning tasks.
- The implementation of some classifiers requires the help of other quantum devices and does not support end-to-end learning from classical data to category labels.

Compared with the related works, the main contributions of this study are summarized as follows:

- The quantum neural network (QNN) model was extended to the facile and scalable variational quantum deep neural network (VQDNN) model that allows an end-to-end training via data preprocessing and a hybrid paradigm of a parametrized quantum circuit.
- Three quantum classifiers were designed based on the proposed VQDNN model for various scales image recognition tasks.
- The angle encoding and amplitude encoding was trained and evaluated, and then were combined in the proposed VQDNN classifiers.
- A series of comparative binary and ten-label classification experiments were conducted using full-size image samples from the MNIST dataset [20] and UCI dataset [2] to demonstrate quantum supremacy.

This paper is structured as follows: In Section 2 the preliminaries of the work are introduced. Section 3 describes the proposed

VQDNN model, three VQDNN based classifiers used in different variations for the experiments, and the comparative experiment between VQDNN and the classical neural network. Section 4 describes the experimental setup and discusses the results. Final conclusions and future works is recommended in Section 5.

2. Preliminaries

2.1. Basic concepts

2.1.1. Quantum bit (Qubit)

The operational architecture of quantum computers is fundamentally different from that of classical computers. The quantum computers adopt quantum bits (qubits) as an operating unit, unlike classical computers adopting binary digits (bits: 0 or 1). Qubits can be 1, 0, and a probabilistic mixture of 1 and 0, called superposition [32]. Thanks to such a key principle, superposition, a quantum computer can conduct specific tasks much faster than classical computers. Generally, discussing a quantum framework via Bra-Ket notation is trivial, where $|\psi\rangle$ and $\langle\psi|$ represent vertical and horizontal quantum state vectors, respectively. A qubit is defined as a linear combination of $|0\rangle$ and $|1\rangle$ which is defined as Eq. (1).

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (1)$$

where α and β are the probability coefficients encoding qubit state information, defined as the square root of the probability to be measured as $|0\rangle$ or $|1\rangle$. Even though qubits can be in both $|0\rangle$ and $|1\rangle$ simultaneously, they are determined as one of $|0\rangle$ or $|1\rangle$ when measured for a definite output.

The unit Bloch Sphere is used to represent a qubit. In the case of $|0\rangle$ and $|1\rangle$, the values are measured across the Z basis. Rotations over the Bloch sphere surface are used to encode and manipulate the data in a pure quantum state. In Eq. (1), the α and β can be considered as the states distances to the state vectors $|0\rangle$ and $|1\rangle$, where a high α indicates being relatively close to $|0\rangle$. Quantum computing can sample the output repeatedly to provide multiple answers for one question.

2.1.2. Quantum data manipulation

To accomplish data transformation and data encoding, a qubit and its quantum state must be manipulated to encapsulate information onto it. Qubits are manipulated through quantum gates, which in turn manipulates the overall quantum state. These gates can allow for complete manipulation over the Bloch sphere, and more specifically complete manipulation of the quantum state vector, which can describe the state of a mixture of more than 1 qubit. The few gates used in this paper are introduced in Eqs. (2)–(5).

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (2)$$

$$R_Y(\theta) = \begin{bmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{bmatrix} \quad (3)$$

$$R_Z(\theta) = \begin{bmatrix} e^{-\frac{i\theta}{2}} & 0 \\ 0 & e^{\frac{i\theta}{2}} \end{bmatrix} \quad (4)$$

$$R_X(\theta) = \begin{bmatrix} \cos(\frac{\theta}{2}) & -i\sin(\frac{\theta}{2}) \\ -i\sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{bmatrix} \quad (5)$$

The accomplished task in Eq. (2) is to entangle two qubits. In Eqs. (3)–(5) a single qubit is allowed to be placed to any position on a Bloch sphere's surface. These gates allow manipulating quantum states.

2.2. Classical data embedding

In the classification circuits, the data samples were first translated from the subject data domain D into the Hilbert space H through a feature map $f: D \rightarrow H$. Various preprocessing methods [41,40,27,15,42] can be integrated into f . In this study, two data encoding patterns were presented, angle encoding and amplitude encoding, adopted in the following parts, which are qubit-efficient or amplitude-efficient for different data dimensions, respectively.

2.2.1. Angle encoding

Angle encoding, sometimes is called qubit encoding or tensor product encoding. Such an employed encoding allows an efficient operation since a constant number of operations in parallel processing is required regardless of the amount of data. However, the number of qubits required depends on the amount of data: one qubit per one component of the input vector. The state preparation scheme requires only a single-qubit to be rotated and thus highly efficient. In contrast, the number of qubits for this encoding is nonoptimal.

All data was normalized into the range $[0, 2\pi]$. Each values x_i was then represented by a single qubit as follows: The rotation operator gates $R_X(\theta)$, $R_Y(\theta)$ and $R_Z(\theta)$ were conducted. As a result, the rotation angle was determined by the data value.

The most commonly used method in some literature is applying the feature map $x_i \rightarrow \cos(\frac{x_i}{2})|0\rangle + \sin(\frac{x_i}{2})|1\rangle$ using the $R_Y(\theta)$ rotation operator as shown as Eq. (6),

$$\begin{aligned} R_Y(x_i)|0\rangle &= e^{(-iYx_i/2)}|0\rangle \\ &= \begin{bmatrix} \cos(\frac{x_i}{2}) & -\sin(\frac{x_i}{2}) \\ \sin(\frac{x_i}{2}) & \cos(\frac{x_i}{2}) \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= \cos(\frac{x_i}{2})|0\rangle + \sin(\frac{x_i}{2})|1\rangle \end{aligned} \quad (6)$$

To map the quantum states encoded from the data points to Hilbert space more dispersed, a slightly more complicated feature map was used as shown in Eq. (7).

$$x_i \rightarrow R_Z(\arctan(x_i^2))R_Y(\arctan x_i)H|0\rangle \quad (7)$$

Given an n -dimensional vector $x = (x_0, x_1, \dots, x_{n-1})$ to be encoded in an n qubit circuit, the encoding operation can be written as Eq. (8).

$$\begin{aligned} U(x) &= R_Z(\arctan(x_0^2))R_Y(\arctan x_0)H \\ &\otimes R_Z(\arctan(x_1^2))R_Y(\arctan x_1)H \otimes \dots \\ &\otimes R_Z(\arctan(x_n^2))R_Y(\arctan x_n)H \end{aligned} \quad (8)$$

2.2.2. Amplitude encoding

Here, given a classical vector, $x = (x_0, x_1, \dots, x_{2^n-1})$, the aim of amplitude encoding was to encode the classical vector into an n -qubit quantum state

$$|\psi\rangle = x_0|00\dots 0\rangle + x_1|00\dots 1\rangle + \dots + x_{2^n-1}|11\dots 1\rangle \quad (9)$$

where the $x_i \in \mathbb{R}$ and the vector $(x_0, x_1, \dots, x_{2^n-1})$ is a normalized vector.

Amplitude encoding has the advantage that it only needs $n = \log(N)$ qubit in encoding an input with N features, so the main reason to choose this encoding scheme is to minimize the number of qubits used, thereby reducing the number of circuit parameters. For example, given a vector with size N , it can be represented with

a $\log_2 N$ qubits system with amplitude encoding. In this work, we consider the input vector with dimension $N = 32 \times 32 = 1024 = 2^{10}$ (for MNIST Dataset) or $N = 8 \times 8 = 64 = 2^6$ (for UCI Digits Dataset) and n -qubit system with or respectively.

2.3. Parameterized quantum circuit

Lewenstein first proposed the concept of quantum perception to apply quantum computers to compute the neural network-like structure [21]. PQC [6] is conceptually similar to a traditional neural network, having trainable parameters in circuits. Like having many layers in a traditional neural network, there are many layers realized by placing the unit layer multiple times in PQC. L times repeated unit layer gives more flexibility to the circuit. Such a deep neural network with multiple layers was proven to obtain outperforming results in many applications.

Let a PQC be denoted as a trainable unitary operation on the n -qubit state, U_θ , applied to a reference state $|\phi\rangle$, which generally equals to $|0\rangle^{\otimes n}$. The trained variable $|\phi_\theta\rangle = U_\theta|\phi\rangle$, Where θ is a vector of a polynomial number of circuit parameters. This circuit layout assumes a unit layer containing single-qubit operations followed by entangling two-qubit operations.

Since an n -qubit quantum neural network encodes a $2^n \times 2^n$ unitary mapping, the number of input qubits is the same as the output qubits. Additionally, small and efficient local gates, such as CNOT, CZ, and rotation gates, were utilized as a building block of the overall unitary. The circuit parameters determined the phases of the rotation gates.

This unit layer of the gate sequence was adopted as a circuit template in this study, as shown in Fig. 2.

3. The proposed VQDNN model and VQDNN classifiers

An end-to-end variational quantum deep neural networks model and the corresponding training algorithm are presented first in this section. Then, three classifiers based on the proposed VQDNN model were designed to perform binary classification tasks. Finally, the classical neural network and VQDNN model was used to perform a comparative experiment of the ten-label classification learning task on the same dataset.

3.1. The end-to-end VQDNN model

Here, PQC was applied as the core to develop a variational quantum model and to realize an end-to-end variational quantum deep

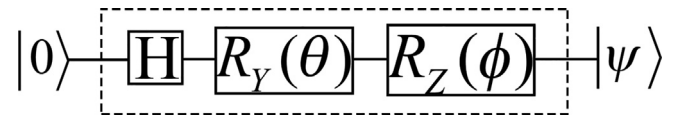


Fig. 1. Angle encoding circuit.

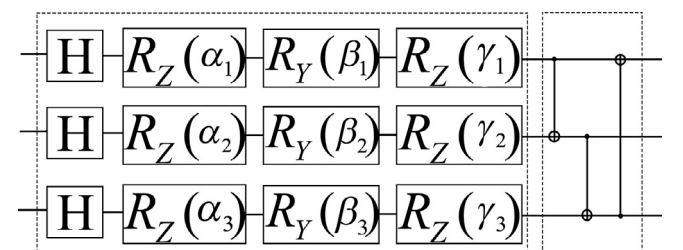


Fig. 2. The unit layer of three qubits for VQDNN model.

neural network. The basic idea of VQDNN was to convert the performed learning task into a minimization problem over the parameters of a quantum circuit, $\vec{\theta}$. The model had five major components: data preparation, data encoding, unitary transformation, entangled state measurement, and classical postprocessing. The first component describes the processing of input data to meet the quantum requirements, using dimensionality reduction, padding, and data flattening methods. After this step, the classical data can be encoded into a quantum state. The next component, data encoding, delineates the skills needed to transform the classical data point to a quantum state, using amplitude encoding or angle encoding schemes. Unitary transformation, the third part of the model, uses H gates, single-qubit rotation gates, and entanglement gates to complete the transformation and entanglement of quantum states. After measuring the quantum state, the expected value of some Hamiltonians is delivered into the postprocessing part. The end-to-end VQDNN model is shown in Fig. 3.

The learning procedure of the VQDNN to solve real-world image classification problems can be summarized as follows:

- The input x of the data point (x, y) was preprocessed and encoded into the input state of the network either as $|\psi\rangle = \sum x_i |i\rangle$ (amplitude encoding) or as $|0\rangle$ followed by a set of rotations $\otimes_{i=0}^{n-1} R(f(x_i))$ (angle encoding), where f is a preprocessing function of x ($\arctan x$ in our research).
- The PQC, represented as $U(\theta)$, was applied to the input quantum state of $|\psi\rangle$ to produce a new entangled state $|\varphi\rangle = U(\theta)|\psi\rangle$.
- The entangled state was observed using a measurement operator. The measurement expected value was used in the classical computation part to produce the predicted label. Define a function, $L(\vec{\theta})$, such that the minimum value of the function, $L(\vec{\theta}^*)$, $\vec{\theta}^* = \arg\min_{\vec{\theta}} (L(\vec{\theta}))$, or the circuit evaluated with the minimized parameters, $U(\vec{\theta}^*)$, corresponds to the solution to some computational problem. The function $F(\vec{\theta})$ can depend on the parameterized circuit in a variety of ways, but the simplest example is one might be where $F(\vec{\theta})$ is the expected value of some Hamiltonian, $\langle \psi(\vec{\theta}) | M | \psi(\vec{\theta}) \rangle$, with $|\psi(\vec{\theta})\rangle$ defined to be the state prepared by parameterized quantum circuit acting on a fixed initial state. Then $L(\vec{\theta})$ may be evaluated by repeatedly preparing $|\psi(\vec{\theta})\rangle$ and measuring M .
- The predicted label $\hat{y}$ and the true label y were used to compute a loss function via classical computation, e.g. the cross entropy loss function $L = \frac{1}{N} \sum_i -[y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)]$, where p_i indicated the probability that the sample i is predicted to be positive.

- Compute $\frac{\partial L}{\partial \theta_i}$ using the hybrid quantum classical automatic differentiation framework [17,23]. Perform gradient descent: $\theta_{k+1} = \theta_k - \eta \frac{\partial L}{\partial \theta_i} \big|_{\theta=\theta_k}$. This procedure was repeated until θ converges, and finally, the trained VQDNN can give the correct classification results.

3.2. Hybrid PCA-VQDNN classifier

In this section, a hybrid PCA-VQDNN classifier was designed based on the above proposed VQDNN model. Principal component analysis (PCA) was employed as the data preprocessing tool aimed at giving the VQDNN model the ability to tackle the common image recognition tasks in high-dimensional space. The end-to-end data flow diagram is shown in Fig. 4. As a baseline, the binary classification tasks were first performed on the hybrid PCA-VQDNN model using the MNIST dataset, where the PCA part serves as the simple feature extractor and the PQC as the classifier.

The learning procedure of the hybrid PCA-VQDNN classifier proceeded from reducing the dimension of the MNIST sample from 784 to 10 and encode it into a quantum state through the angle encoding method, which is defined in Fig. 1. The core component of the quantum classifier is shown in Fig. 2, which contains all trainable parameters in the quantum circuit and can be stacked in multiple layers. After the unitary evolution of PQC, mainly using the Hadamard gate, single-qubit rotation gate (R_y and R_z), and entanglement gate (CNOT) to complete the transformation and entanglement of quantum states, the output quantum state enters the subsequent Pauli-Z measurement circuit. The classical postprocessing function, employing the cross-entropy loss function here in this research, was applied to the measurement result to minimize the loss function. The quantum circuit and the classical optimizer work alternately until the optimal parameters were found.

3.3. Amplitude encoding based VQDNN classifier

In the PCA-VQDNN classifier, to reduce the desired number of qubits, classical principal component analysis was adopted to preprocess the data sample. However, here, how the proposed VQDNN model performs on the original datasets was the objective. Fig. 5 shows that VQDNN uses the amplitude encoding method to encode the MNIST or UCI handwritten digit sample to ten or 6-qubit quantum systems, respectively.

If the number of pixels in an original image was not exactly equal to a n -th power of 2, the image can extend by padding the input vector to reach 2^n -dimensional.

For MNIST, the 784-dimensional input was padded into a 1024-dimensional normalized vector $\bar{x}$, which can be converted into an amplitude-encoded 10-qubit state denoted by $|\varphi\rangle$. The subsequent classification circuit is shown in Fig. 5.

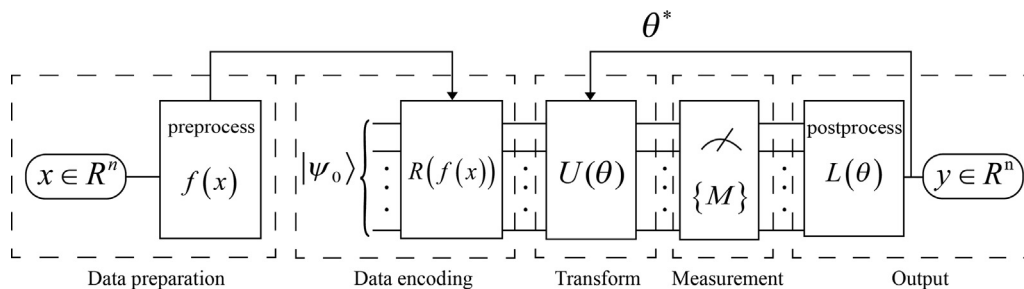


Fig. 3. The schematic diagram of end-to-end VQDNN model.

3.4. Hybrid encoding based VQDNN classifier

In the above two parts, classifiers were designed to explore the performance of the proposed VQDNN model in low-dimensional feature space and high-dimensional feature space, which were

implemented using angle rotational encoding or amplitude encoding techniques, respectively. If the two architectures were mixed together, is the proposed VQDNN model still effective? The first part of the classifier uses amplitude encoding techniques to achieve exponential coding supremacy, and the subsequent circuit

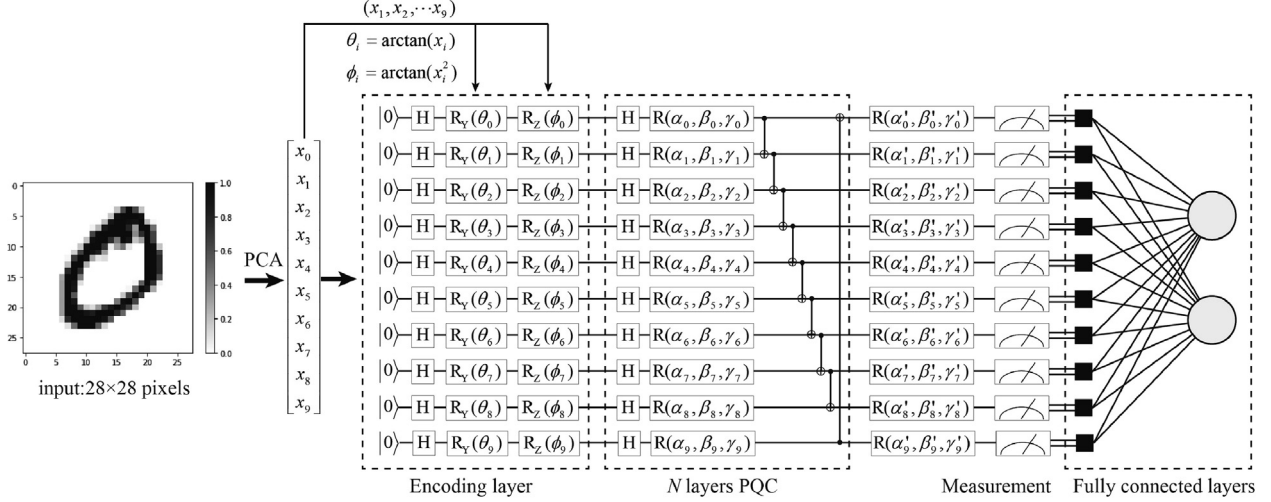


Fig. 4. The diagram of the hybrid PCA-VQDNN classifier. The first part is data preprocessing. The middle part is the data encoding circuit. By going through one or more layers of PQC, one measures all ten qubits, and ultimately takes the expected value for classic postprocessing.

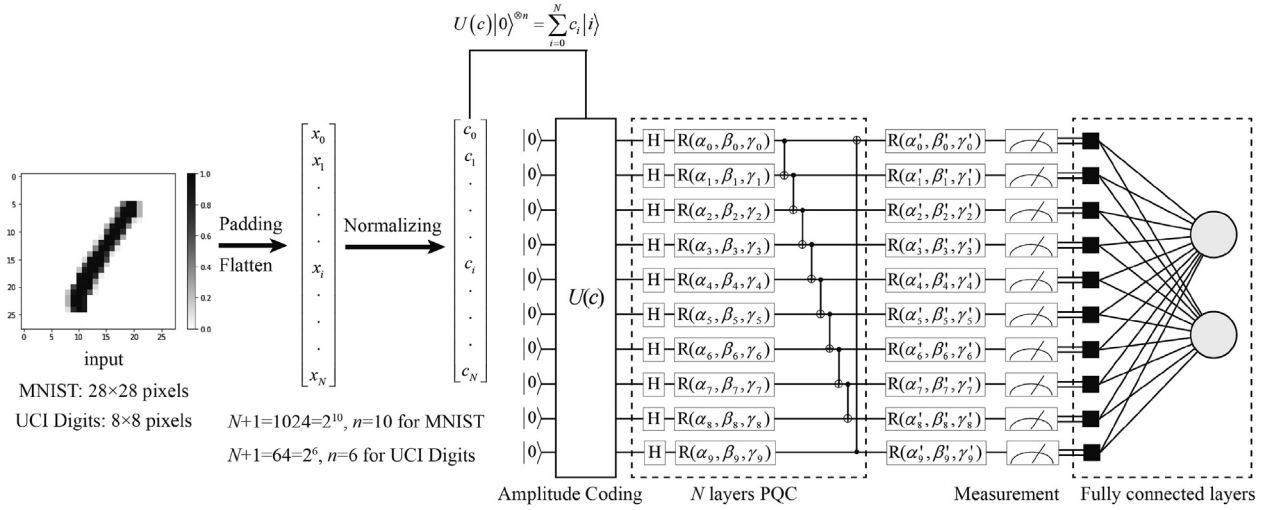


Fig. 5. Diagram of the amplitude encoding-based VQDNN classifier. The first part is padding and flattening the input samples. $U(c)$ indicates amplitude encoding circuit. Subsequently, N layers PQC is applied to the input state. At last measure all the ten qubits and take the expected value for classic post-processing.

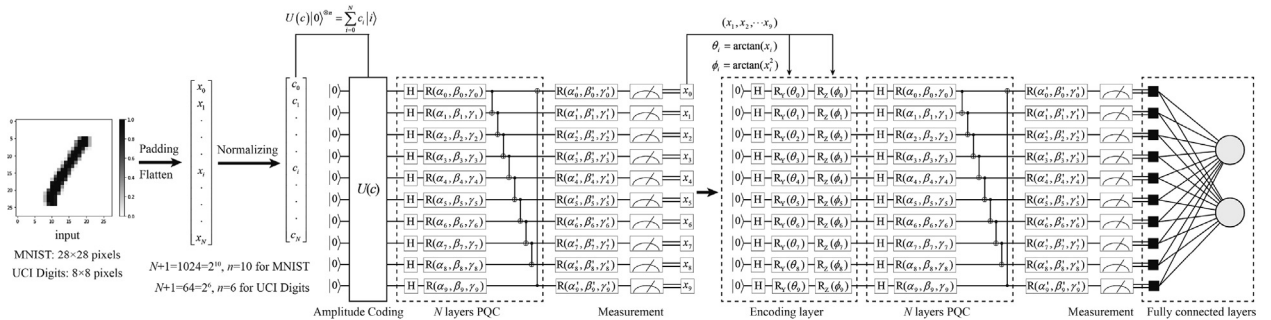


Fig. 6. Diagram of the hybrid encoding-based VQDNN classifier. The first part is amplitude encoding. After passing through the first PQC and measuring, the expectations are encoded into angles of qubits in the second PQC. The measurement results are used in classical postprocessing.

uses angle encoding to span the data to the large Hilbert space. The proposed VQDNN model's performance in a sophisticated network structure was the objective.

For large-size learning samples, amplitude encoding was used to encode them. After passing through the first PQC and measuring, the expectations were encoded into angles of qubits in the second PQC. The measurement results were used in classical postprocessing. The whole process is shown in Fig. 6. Generally, the procedure consists of the following six steps:

- After padding and flattening the input samples, the classical data can be encoded into a quantum state using the amplitude encoding method.
- The input quantum state was transformed and entangled through the first PQC, which can be stacked in multiple layers.
- All the qubits of the first PQC were measured in Pauli-Z before passing the expected values into the second PQC using the angle encoding method.
- Similar to the first PQC, the input quantum state was transformed and entangled in the second PQC, which can be stacked in multiple layers.
- After measuring and passing the expected values, the loss function can be calculated from the predicted value and the true value.
- The tunable parameters in the two PQCs and the classical fully connected layers were optimized using a classical optimizer.

3.5. Benchmark design of VQDNN model

In the above four parts, an end-to-end VQDNN model was proposed and three classifiers were designed to explore the performance of VQDNN in low-dimensional feature space, high-dimensional feature space and sophisticated network structure, which were implemented using PCA, amplitude encoding and hybrid encoding technology, respectively. All three classifiers performed well on the learning task. The next question naturally was how the performance of the VQDNN model-based classifier compared with the classical original neural network. Under the limitation of the same number of trainable parameters in the classical classifier, does the VQDNN classifier still have satisfactory classification performance?

Here, a ten-label classification task benchmark was conducted on the UCI digits dataset. In the quantum setting, shown in Fig. 7, the UCI digits image was first flattened. After normalization, the vector was encoded into a 6-qubit quantum state using the amplitude encoding method. Then, a multilayer PQC was applied to the input state. After measurement, the expected values were injected into a 6×10 fully connected layers for classical postpro-

cessing. In the classical neural network, the simplest network structure was used that contains just one hidden layer, as shown in Fig. 8. In the comparative setting, we control the number of tunable parameters by adjusting the number of PQC layers and nodes in the hidden layer in the classical network, and thus analyze the relative performance of the VQDNN and classical neural network on classification tasks.

4. Numerical simulation & results discussion

In this section, four numerical simulation experiments are implemented. The codes were written purely in Julia [7] and the quantum circuit simulation was performed by the open-source variational quantum circuit simulator VQC [24]. Adaptive moment estimation (Adam) was used in the parameter updating.

4.1. VQDNN was effective in low feature spaces

VQDNN was first trained using hybrid PCA-VQDNN architecture on the MNIST dataset. Each example of MNIST was described by 784 features, which required too many Qubits for some NISQ processors. Here, PCA was adopted to reduce the 784 features to 10 features. After the reduction, the 10 features were encoded into the quantum states using rotational encoding and transformed by shallow PQC. Finally, the measurement results were used to calculate the loss value and predict the class label.

In this numerical simulation, two binary datasets were extracted including '0, 1' and '3, 8' samples only from the original

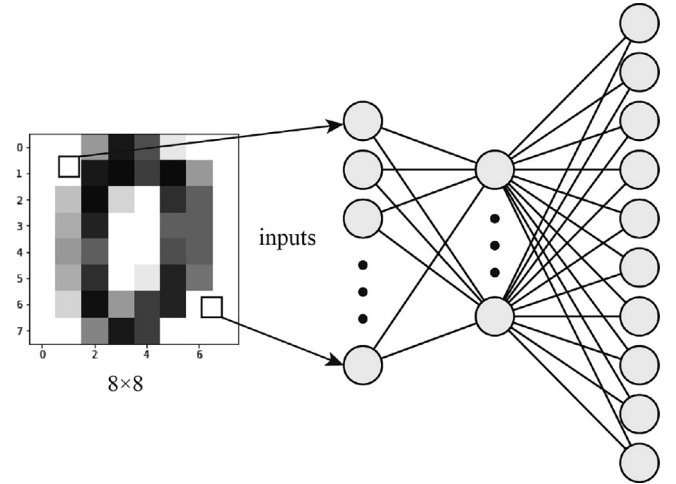


Fig. 8. Ten-label classification learning task of the classical neural network.

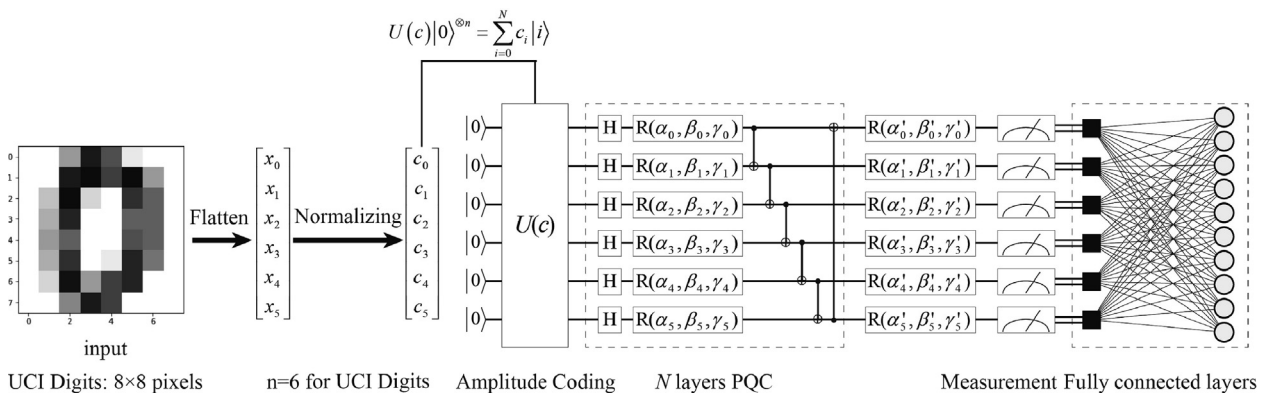


Fig. 7. Ten-label classification learning task of VQDNN.

data. For each class, 1000 samples were used as the training data and 200 samples were set as to the test to verify the accuracy of the algorithm. A stochastic gradient optimizer called adaptive moment estimation (Adam) was used with a learning rate of 0.01. To reduce the randomness of the experiment, Each experiment was conducted 10 times, and the average of 10 times was taken in each iteration as the accuracy value and loss value of this iteration. The test accuracy and loss curves of single-layer PQC, double-layer PQC and triple-layer PQC with 80, 110, and 140 parameters in the 0, 1 classification and 3, 8 classification are shown in Fig. 9. It can be seen that the VQDNN model had achieved better classification accuracy on the 0 and 1 binary classification tasks than on the 3 and 8 binary classification tasks. Observing the same classification task, as the number of PQC layers increases, the classification performance of the VQDNN model was significantly improved. The test accuracies of single-layer PQC, double-layer PQC, and three-layer PQC all reached 99% under the two-class classification task of 0 and 1. However, it is worth noting that as the number of PQC layers increased, the classifier required fewer iterations to achieve 99% classification accuracy. Single-layer PQC and double-layer PQC can achieve 99% classification accuracy after 60 iterations, which was significantly better than single-layer PQC after 90 iterations.

To visualize the training performance, the training loss is shown on the right side of Fig. 9. As the number of training iterations increased, the losses of the three PQCs will gradually decrease under the two classification tasks. However, after carefully checking the loss curve, it was found that the two-layer and three-layer PQCs decreased significantly faster on the 0 and 1 classification tasks, and finally reached approximately 0.01. In summary, it can be seen that the hybrid PCA-VQDNN algorithm exhibits excellent performance in the low-dimensional feature space with limited quantum computing resources. By increasing the number of trainable parameters (increasing the number of PQC layers), the model will converge faster, and the training loss can be reduced to a minimum faster.

4.2. VQDNN was effective in high-dimensional spaces

To explore the classification performance of VQDNN in high-dimensional feature spaces, here we implement the binary classification learning task of number '0, 1' and '3, 8' on UCI handwritten digits dataset and MNIST dataset, respectively.

4.2.1. UCI hand-written digit data

A: '0, 1' vs. '3, 8' classification.

First, we trained VQDNN on the UCI hand-written digit dataset. Filtering out the digits '0, 1' and '3, 8' in the UCI handwritten digits dataset, there were 765 training samples and 360 testing samples left for '0, 1'. The numbers '3' and '8' were also screened out, and 769 training samples and 357 test samples were obtained. In this experiment, we first flattened the 8×8 images to the 64×1 column vector. After normalization, the vector was encoded into a 6-qubit quantum state using the amplitude encoding method. Then, a multilayer PQC is applied to the input state.

Depending on the number of layers included in the PQC, each VQDNN contains a different number of trainable parameters. The number of different parameters determines the learning performance of the quantum classifier. In this numerical simulation experiment, we set the number of layer to 1–5 to explore the performance of the VQDNN under different circuit depths. Adam was used as the optimizer with a learning rate of 0.01. To reduce the randomness of the experiment, we repeated each quantum experiment 10 times, and took the average of 10 values as the value of the current iteration. We trained each classifier in the 0, 1 and 3, 8 binary classification tasks 100 times, and the results are shown in Fig. 10 and Fig. 11.

In Fig. 10, when the quantum classifier takes a single layer, there were 48 trainable parameters in the quantum neural network ($6 \times 3 + 6 \times 3 + 6 \times 2$). At the end of the training process, the average test accuracy for 10 runs was approximately 0.9747. This means that for 360 test samples, the quantum classifier made 9 errors. As the number of PQC layers increased to 5, the number of trainable parameters reached 120. At the end of the training process, the average test accuracy of 10 runs is approximately 0.9989, and the quantum classifier can almost completely classify correctly. It can also be seen from Fig. 10 that as the depth of the quantum circuit increases, the classifier can achieve better classification accuracy, and it can also converge faster. However, from the fact that the second half of the accuracy curves at depths 4 and 5 almost overlap, it can be seen that the classification accuracy and convergence speed cannot be improved all the time. With the increase in trainable parameters, the network has been saturated.

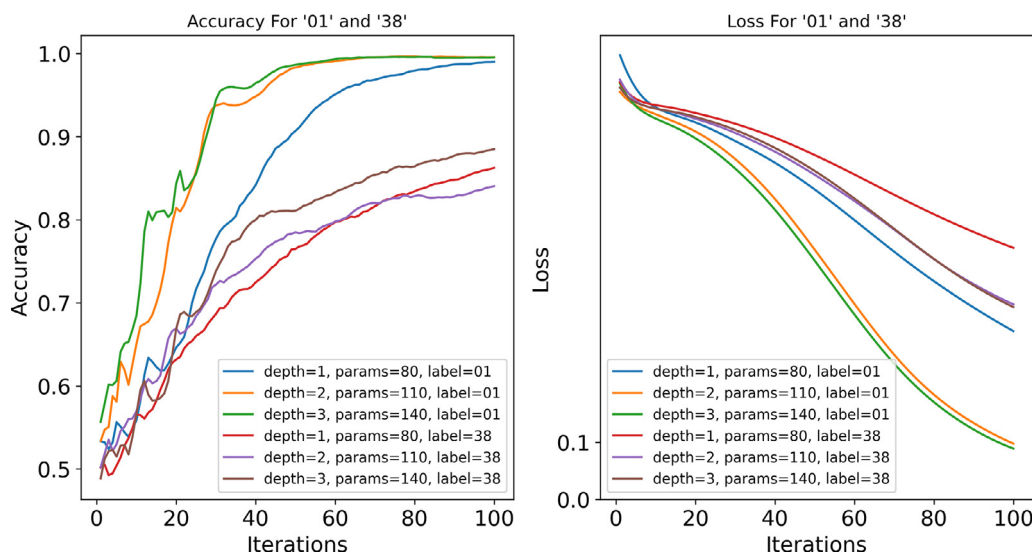


Fig. 9. Illustration of hybrid PCA-VQDNN architecture single-layer PQC, two-layer PQC and three-layer PQC on the 0, 1, and 3, 8 binary classification tasks test accuracy and training loss curve with the number of iterations. As the number of training iterations increases, the accuracy of PQC with more layers is higher than that of single-layer PQC, and the loss also drops faster.

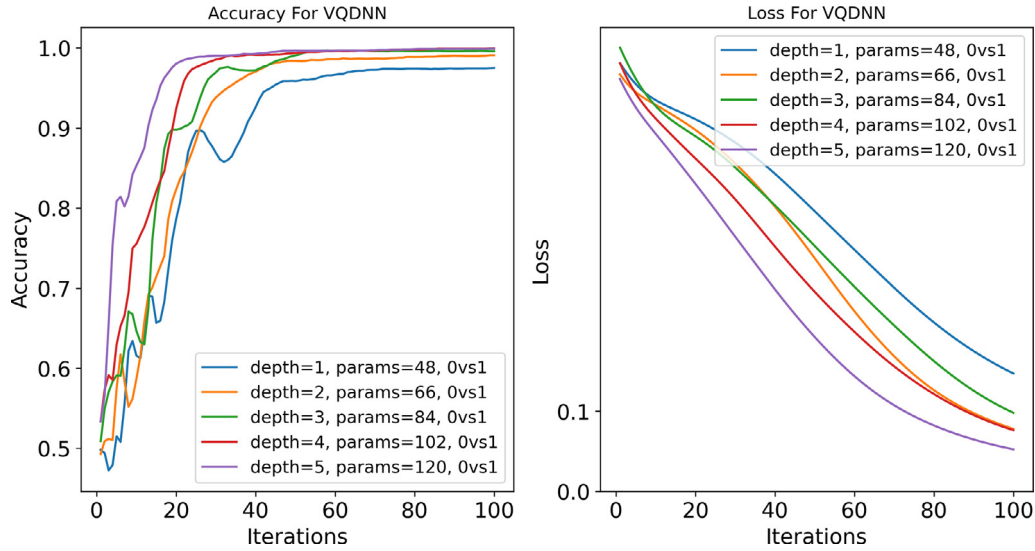


Fig. 10. The test accuracy and training loss of the 1–5 layer PQC using the VQDNN architecture on the 0, 1 classification task vary with the number of iterations.

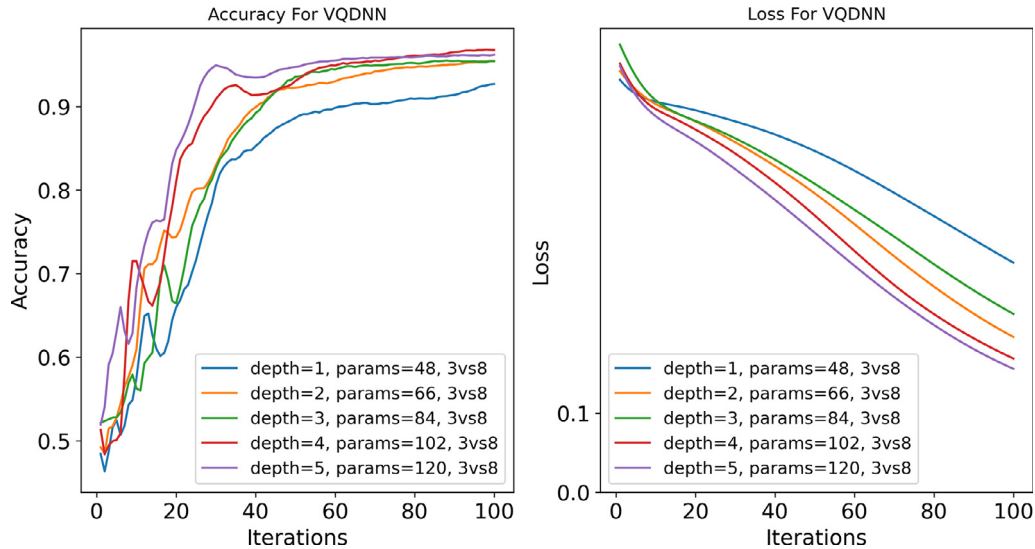


Fig. 11. The test accuracy and training loss of the 1–5 layer PQC using the VQDNN architecture on the 3 and 8 classification task vary with the number of iterations.

It can be seen from the loss curve on the right side of Fig. 10 that as the depth of the quantum circuit increases, the training loss converges faster and faster.

Fig. 11 shows the test accuracy and training loss of PQC on 3 and 8 classification tasks as a function of the number of iterations. Similar to the 0, 1 classification task, in this experiment, as the depth of the quantum circuit increases, the classifier can achieve better classification accuracy and converge faster. This was consistent with the performance on the 0, 1 classification task.

To compare the performance difference of the VQDNN model on different classification tasks, we took the change curves of the PQC with depths 1, 3, and 5 on the two classification tasks of '0, 1' and '3, 8' to draw Fig. 12. It can be seen from the figure that the PQC of the same depth can achieve better accuracy on 0 and 1 classification tasks and converge faster. This was because 3 and 8 classification tasks were more complicated than 0 and 1, and achieving similar classification accuracy requires more trainable parameters and more iterations.

B: quantum vs. classical classification.

To compare with the classic neural network, we used a single hidden layer classic fully connected neural network to perform the same two-class learning task. The classic neural network uses the same 64 pixel values as input, was set as a single hidden layer, the output layer had 2 nodes, and the hyperparameter settings were exactly the same as VQDNN. This classic benchmark was designed to limit the number of parameters to make a fair comparison with the quantum model. The experimental results are shown in Fig. 13 and Fig. 14. It can be seen from the two figures that the classification performance of VQDNN was significantly better than that of the classic neural network under the limitation of the number of similar parameters, regardless of whether it was a 0, 1 classification or a 3, 8 classification.

4.2.2. MNIST

To fully explore the classification performance of VQDNN in high-dimensional spaces, we trained VQDNN on the MNIST dataset

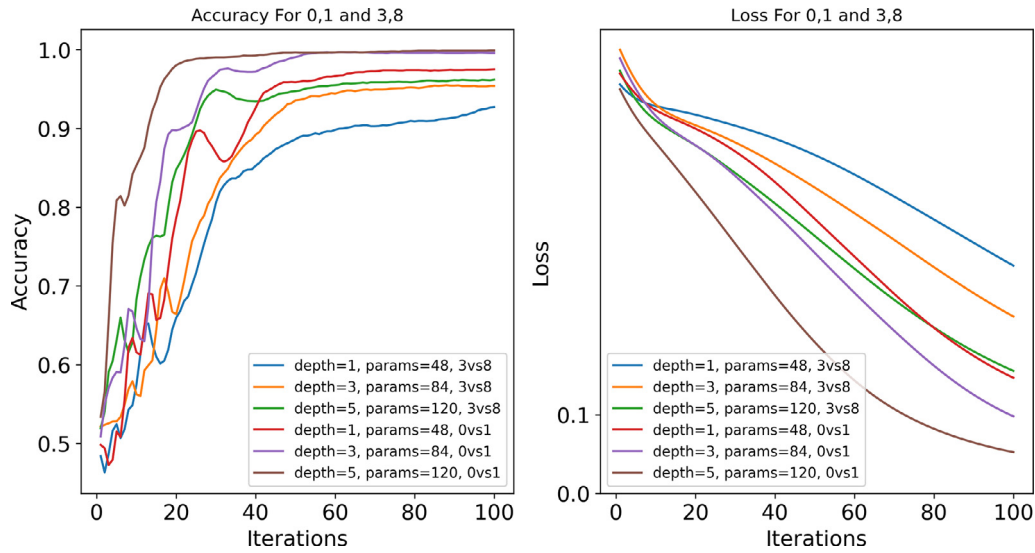


Fig. 12. Comparison of the test accuracy and training loss with the number of iterations on the 0, 1 and 3, 8 classification tasks of the 1–5 layer PQC using the VQDNN architecture.

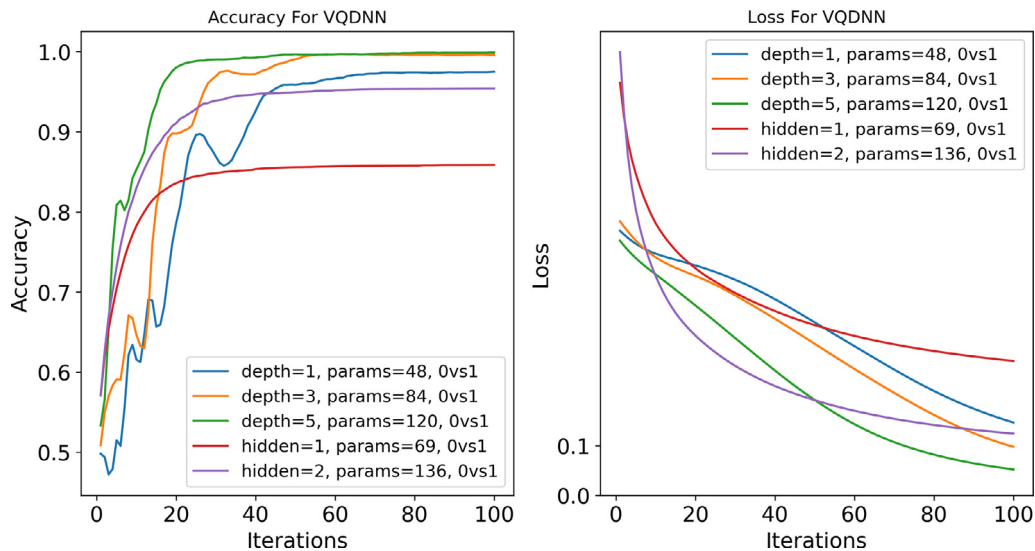


Fig. 13. The test accuracy and training loss of the quantum classifier and the single hidden layer classical neural network classifier on the 0, 1 classification task are compared with the change curve of the number of iterations.

which was more difficult for the currently available quantum computing resources. Because the 784-dimensional MNIST sample was still too large for the current quantum processor, the previous research literature mostly downsampled the 28×28 digital image to 8×8 or 4×4 . However here, we padded the 28×28 images to 32×32 , taking zero for the outermost pixels of the image and adopt amplitude encoding method. Therefore, the 1024-dimensional pixel value was encoded into a 10-qubits quantum system. We extracted two binary datasets including '0, 1' and '3, 8' samples only from the original dataset. For each class, we used 1000 samples as the training data and 200 samples as the test set to verify the accuracy of the classifier. We trained each classifier in the 0, 1 and 3, 8 binary classification tasks 10 times to take the average. The result is shown in Fig. 15.

For a one-layer quantum classifier, there were 80 trainable parameters in the VQDNN ($10 \times 3 \times 1 + 10 \times 3 + 10 \times 2$). As the PQC depth increases to 2, 3, and 5 layers, the trainable parameters were 110, 140, and 200 respectively. Adam was used as the opti-

mizer with a learning rate of 0.05. It can also be seen from Fig. 15 that in the two classification tasks, as the depth of the quantum circuit increased, the classifier can achieve better classification accuracy and converge faster. It can also be observed that the PQC of the same depth can achieve better accuracy on 0 and 1 classification tasks and converge faster. This was because 3 and 8 classification tasks were more complicated than 0 and 1, and achieving similar classification accuracy requires more trainable parameters and more iterations.

According to the results of two numerical simulation experiments, it can be concluded that the proposed VQDNN could also complete the two-class learning task well in the high-dimensional feature spaces.

4.3. VQDNN was effective in sophisticated network structure

According to the two research results above, we know that VQDNN can complete the two-class learning task well in both

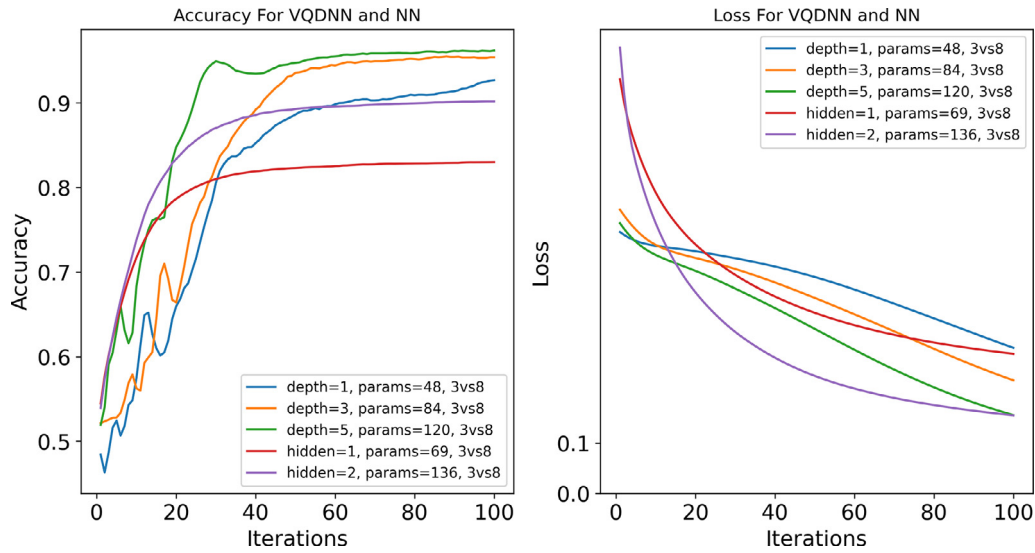


Fig. 14. The test accuracy and training loss of the quantum classifier and the single hidden layer classical neural network classifier on the 3, 8 classification task are compared with the change curve of the number of iterations.

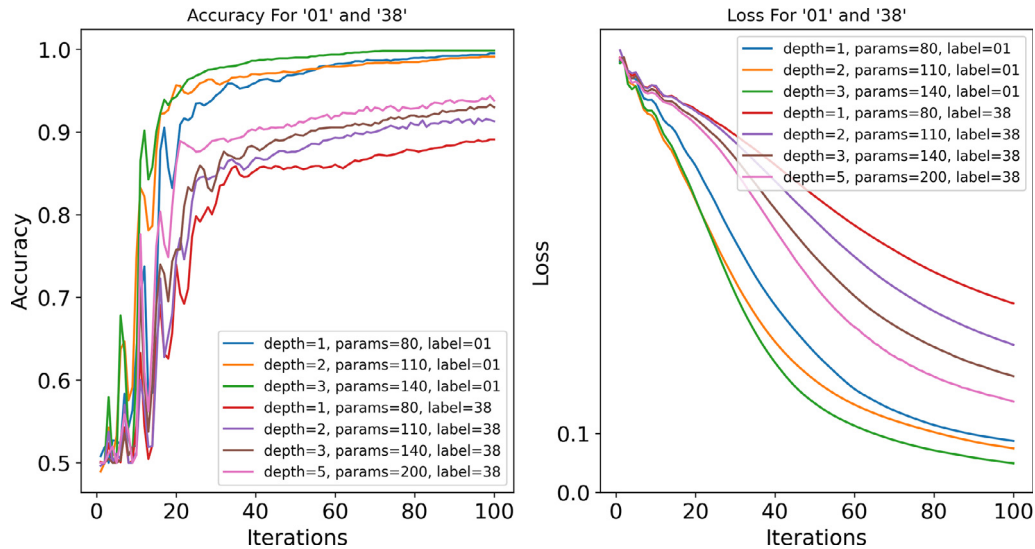


Fig. 15. The binary classification accuracy and loss curve of the multilayer PQC using the VQDNN architecture varies with the number of iterations.

low-dimensional feature spaces and high-dimensional feature spaces. If we mixed the two architectures above together, does the proposed VQDNN model still be effective? Below, we will verify the binary classification performance of the model on the two datasets.

4.3.1. UCI hand-written digit data

We first encoded the 64-feature digit image into a 6 qubits quantum system using the amplitude encoding method. After the unitary transformation of the first PQC, we measured all 6 qubits. Passed the measured expected values into the second PQC, using the rotational encoding method. After unitary transformation, 6 qubits were measured to obtain 6 expected values, followed by a 6×2 fully connected neural network, and finally the classification value is given.

A: 0, 1 binary classification task

For the depth = 1 quantum classifier, where the first PQC and the second PQC all take one-layer circuit, there are 84 trainable parameters in the quantum neural network ($6 \times 3 \times 1 + 6 \times 3 \times 6 \times 3 \times 1 + 6 \times 3 + 6 \times 2$). Adam was used as

the optimizer with a learning rate of 0.05. At the end of 100 training epochs, the test accuracy was approximately 0.9975. This means for 360 testing samples, the quantum classifier made one mistake. As increase the quantum classifier depth to 2, the trainable parameters reached 120. As shown in Fig. 16, almost from the 30th epoch, the test accuracy had been stable at 1.00. It can also be seen that as the depth of the quantum circuit increased, the training loss converged quickly.

Model performances of test accuracy (left) and training loss (right) modified VQDNN depth. In both figures, the curve depth takes 2 and parameters takes 120 is the better one.

B: Amplitude encoding versus Hybrid encoding

To analyze the classification performance of the VQDNN model under amplitude encoding and hybrid encoding, we conducted benchmark experiments, and the results are shown in Fig. 17 and Fig. 18.

It can be seen in Fig. 17 that the classification performance of the VQDNN model in the case of amplitude encoding and hybrid encoding in the 0, 1 classification task is significantly better than the classical neural network. Comparing Amplitude coding (Ex2)

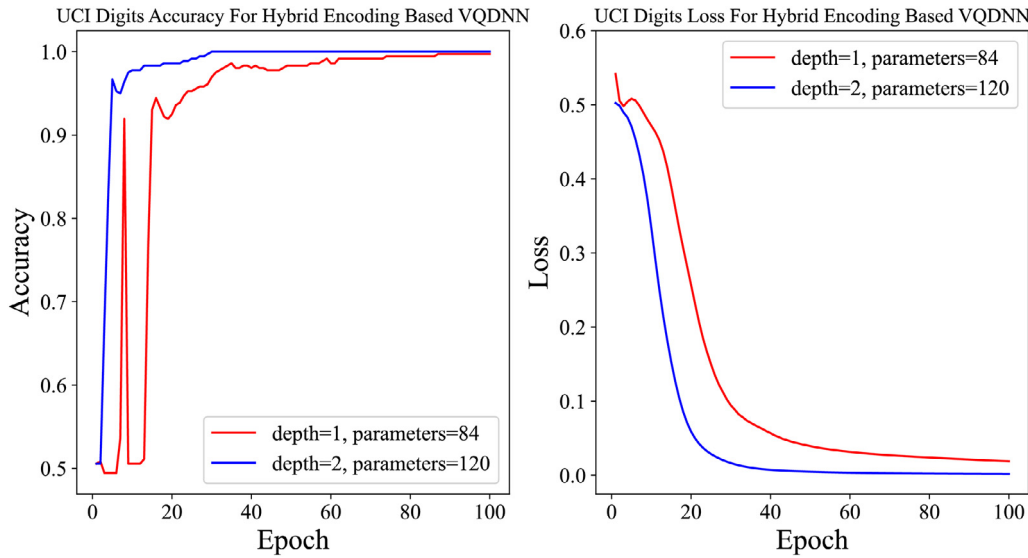


Fig. 16. Model performances of test accuracy (left) and training loss (right) modified VQDNN depth. In both figures, the curve depth is 2 and parameters takes 120 is the better one.

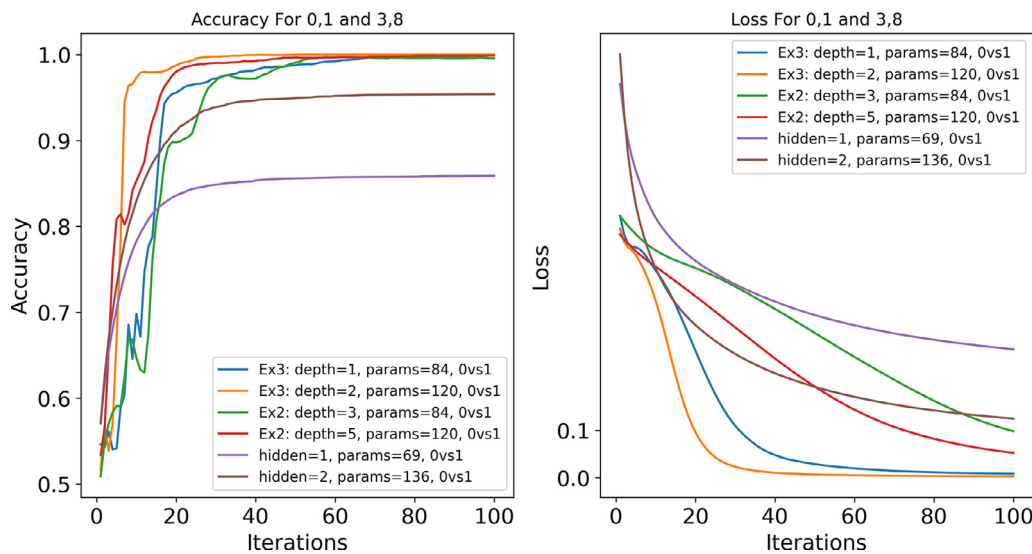


Fig. 17. The accuracy and loss of the VQDNN model based on amplitude encoding and hybrid encoding and the classical neural network on the 0, 1 binary classification learning tasks vary with the number of iterations.

and hybrid coding (Ex3), it can be seen that under the same number of trainable parameters, the classification performance of VQDNN in complex networks was not worse than the simple network structure under Amplitude encoding in the previous section. The VQDNN model in the 3 and 8 classification tasks in Fig. 18 also had a similar performance. This shows that our proposed VQDNN model was effective in complex network structures.

4.3.2. MNIST

A: 0, 1 binary classification task.

After padding the 28×28 MNIST images to 32×32 and encoding the 1024-dimensional pixel value into a 10-qubits quantum system, we applied the unitary transformation formed by the first PQC. After measuring and taking the expected value of 10 qubits, we used the rotation encoding method to transport the expected value to the second PQC. After unitary transformation, all qubits were measured, and the expected value was used for binary classification.

For each class, we took 1000 samples as the training data and 200 samples as the testing data to train the quantum classifier. For the depth = 1 quantum classifier, where the first PQC and the second PQC all take a one-layer circuit, there were 140 trainable parameters in the quantum neural network ($10 \times 3 \times 1 + 10 \times 3 + 10 \times 3 \times 1 + 10 \times 3 + 10 \times 2$). Adam was used as the optimizer with a learning rate of 0.05. At the end of 100 training epochs, the test accuracy was approximately 0.9975. This means that for 400 testing samples, the quantum classifier made one mistake. by increasing the quantum classifier depth to 2, the trainable parameters reached 200. As seen from Fig. 19, almost from the 48th epoch, the test accuracy was stable at 1.00. It can also be seen that as the depth of the quantum circuit increases, the training loss converged quickly.

Model performances of test accuracy (left) and training loss (right) modified VQDNN depth. In both figures, the curve depth takes 2 and parameters takes 120 is the better one.

B: quantum vs. classical classification

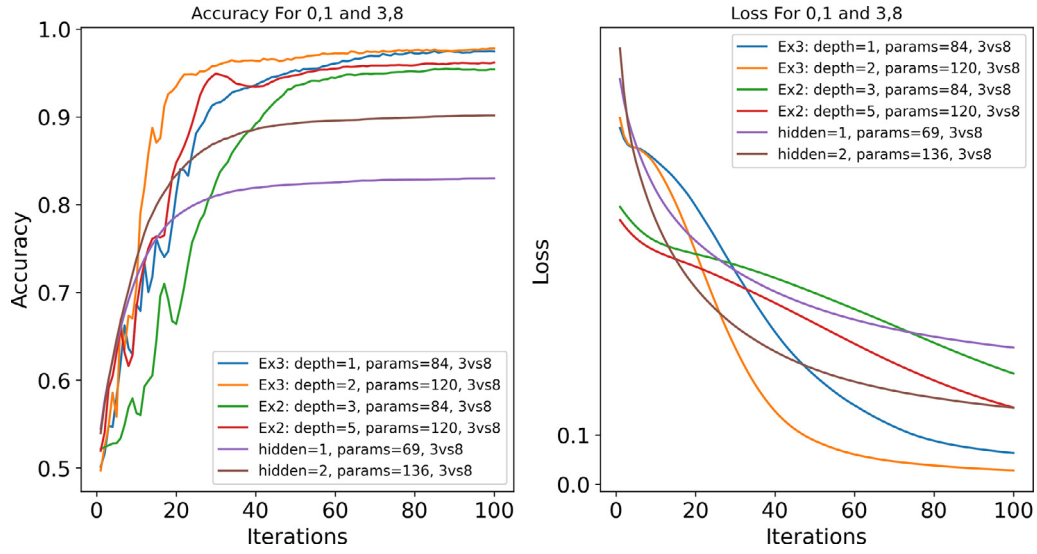


Fig. 18. The accuracy and loss of the VQDNN model based on amplitude encoding and hybrid encoding and the classical neural network on the 3, 8 binary classification learning tasks vary with the number of iterations.

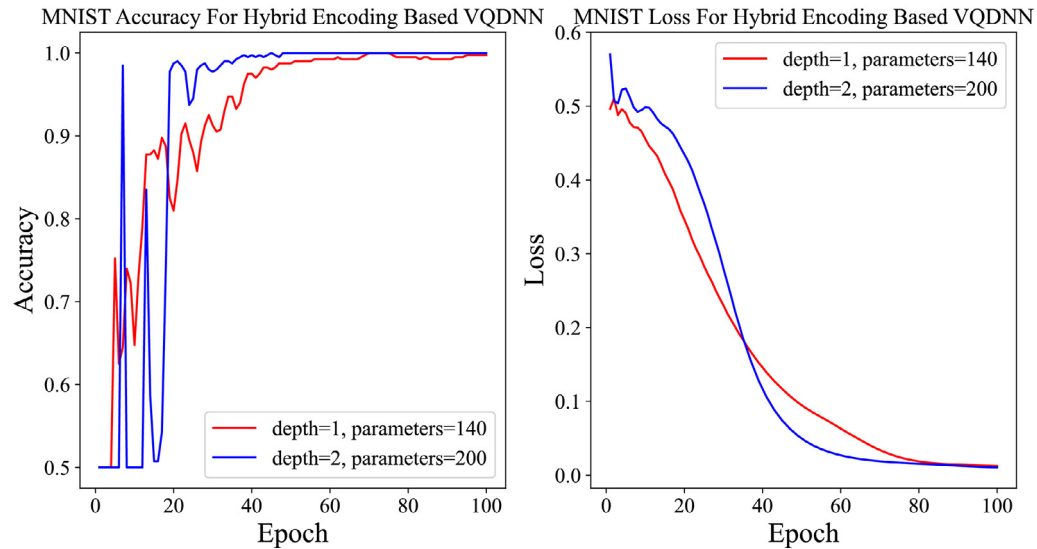


Fig. 19. Model performances of test accuracy (left) and training loss (right) modified VQDNN depth. In both figures, the curve depth takes 2 and parameters takes 200 is the better one.

To explore the performance difference of VQDNN-based classifiers for classification tasks of different degrees of difficulty under complex networks, we further extracted a binary classification data set containing '3' and '8' samples from the original dataset. Similarly, for each category, we used 1000 samples as the training data and 200 samples as the test set to verify the performance of the classifier. To more accurately reflect the performance of the classifier, we iteratively trained each classifier in the 0, 1, 3, and 8 binary classification tasks, and ran it 10 times to average. The results are the four curves identified by Ex3 in Fig. 4.20 Shown.

To compare the performance differences of the classifier based on the VQDNN model in the case of amplitude coding and hybrid coding, we also plotted the VQDNN classifier data based on the amplitude coding in the previous section in Fig. 20, that is, the three curves identified by Ex2.

Regardless of 0, 1 classification or 3, 8 classification, with the increase of PQC depth, the classification performance was improved. For the 0, 1 classification task, when the PQC contains 140 trainable parameters, the classification performance of the

complex network was slightly better than that of the pure amplitude coding classifier, as shown in the blue and red curves. For 3 and 8 classification tasks, the classification performance of the pure amplitude coding classifier was slightly better than the classification performance of the complex network. It can also be seen that it was not possible to simply judge which of these two network structures was better or worse. In actual use, you should make careful judgments for different learning tasks. However, it can be seen in Fig. 21 that although the trainable parameters of the two classifiers were an order of magnitude worse than the classic fully connected network, the performance was similar.

4.4. Compared with the classical neural network, VQDNN has better performance on ten-label classification tasks

4.4.1. quantum vs. classical classification task on UCI datasets

Here we conducted a ten-label classification learning task comparative benchmark experiment on the UCI digit dataset. The

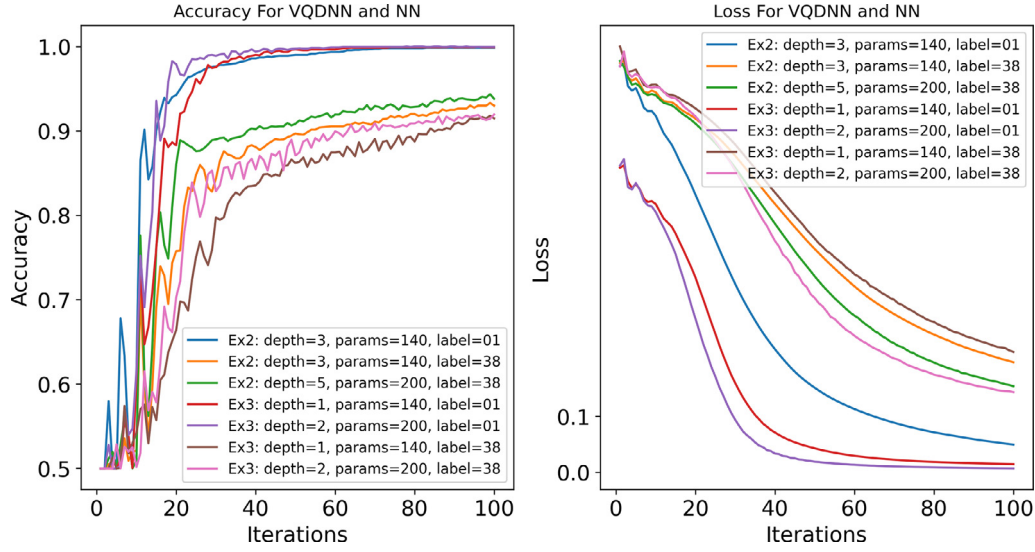


Fig. 20. The accuracy and loss of the binary classification of multilayer PQC under the amplitude encoding and hybrid encoding of the VQDNN architecture vary with the number of iterations.

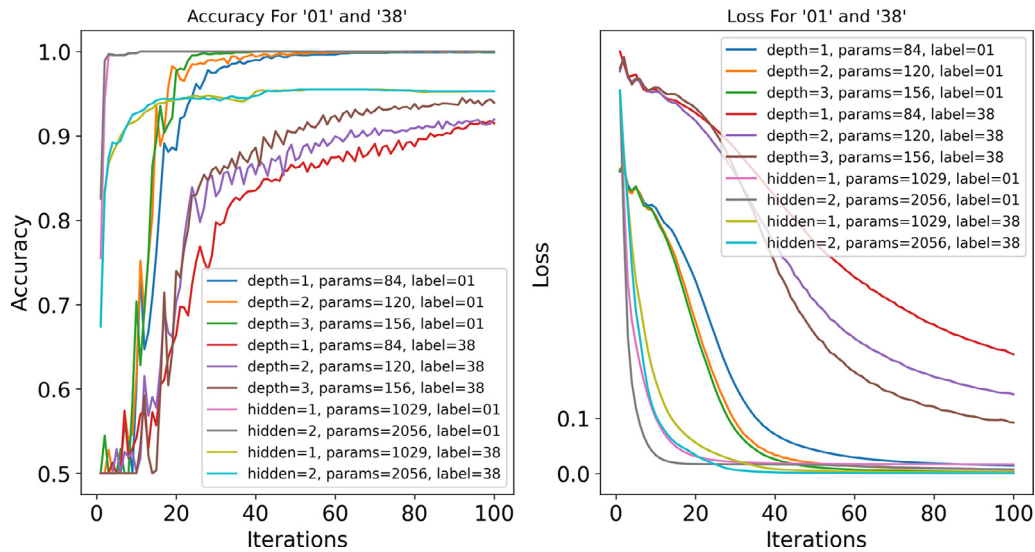


Fig. 21. The binary classification accuracy and loss curve of the hybrid encoding classifier using the VQDNN architecture and the classical fully connected neural network vary with the number of iterations.

VQDNN classifier used amplitude encoding to compress the 64-dimensional vectors into a 6-qubit quantum system.

The classical neural network used the same 64 pixels as an input. The hidden layer has only one node and the output layer had ten nodes. The design for this classical benchmark aims to limit the number of parameters for a fair comparison to the quantum model.

The PQC had 6 qubits input, one layer of unitary transform, and 6 qubits output (Fig. 7). Each qubit had a tunable unitary transform per layer, so there were 96 parameters ($6 \times 3 \times 1 + 6 \times 3 + 6 \times 10$) in the circuit. Adam was used as the optimizer with a learning rate of 0.05.

Fig. 22 shows the variation curve of test accuracy and training loss with the number of iterations during the training process. Look at the right of the figure first.

When the hidden layer has 4 nodes, the classical classifier had 310 parameters. The VQDNN classifier had 168 parameters when the PQC had 5 layers of unitary transform. Importantly, roughly

only half (exactly 54.19%) of the number of parameters associated with the analogous classical neural network (Fig. 8) were used for the same classification task. However, the test accuracy that the two models can achieve after 200 epochs of training was basically the same. To be more precise, VQDNN had a slight advantage.

The same phenomenon can also be seen in the comparison between the left and the middle of Fig. 22. In the left figure, when the trainable parameter is approximately 90, the classification accuracy of VQDNN far exceeds the classification accuracy of the classic neural network. The middle figure shows the change in accuracy when the trainable parameter is approximately 160. It can also be seen from the figure that the classification accuracy of the classic neural network was still quite large compared to VQDNN.

It can be seen in Fig. 23 that whether it as classical or quantum, as the depth (the number of hidden layer nodes) increased, the faster the loss decreased.

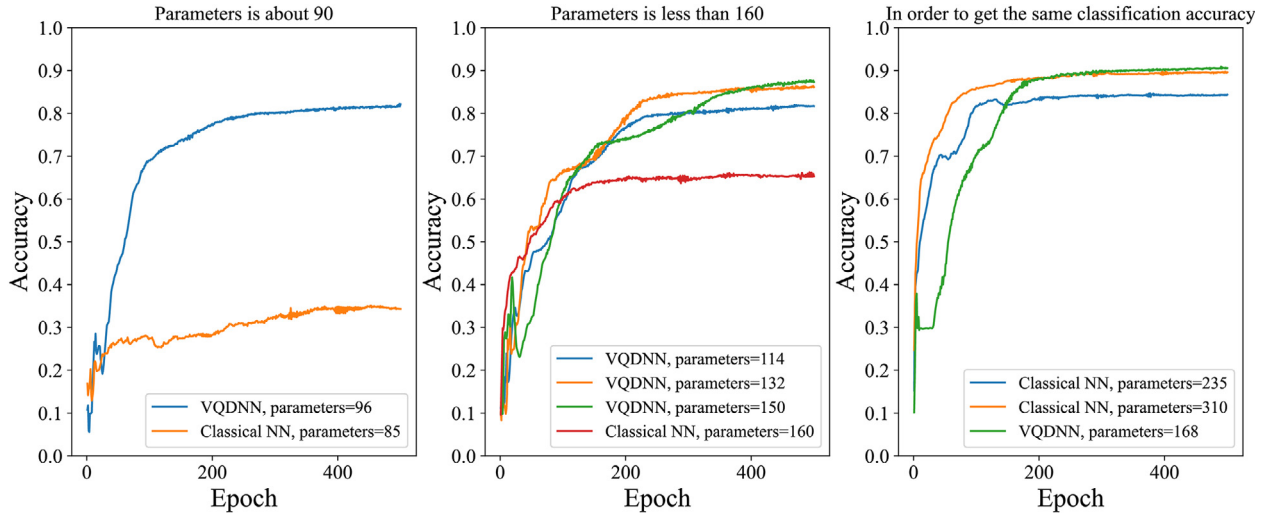


Fig. 22. Model performances of test accuracy of VQDNN and Classical Neural Network modified the model depth. In the left figure, the layer depth is set to 1 and both the number of parameters is about 90. VQDNN significantly excels classical networks. In the middle figure, VQDNN's leading advantage can still be maintained as the depth increases. In the right figure, VQDNN can achieve better classification accuracy using only 54.19% of the parameters in the classical network.

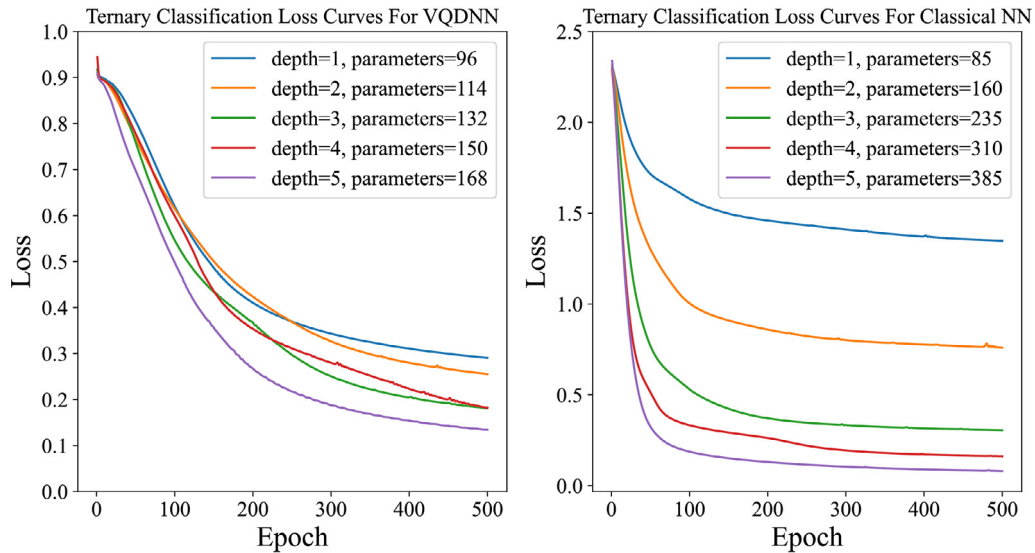


Fig. 23. Model performances of training loss of VQDNN (left) and classical neural networks (right) modified VQDNN depth. In both figures, with increasing depth, the loss declines faster.

4.4.2. quantum vs. classical classification task on MNIST datasets

Here we still padded the 28×28 images to 32×32 , take zero for the outermost pixels of the image and adopt the amplitude encoding method. Therefore, the 1024-dimensional pixel value is encoded into a 10 qubit quantum system. Considering that a ten-qubit simulation system was time-consuming, for each category, we used 200 samples as training data and 100 samples as a test set to verify the performance of the classifier. We train each classifier iteratively 500 times. For the classic fully connected network, because the number of nodes in the hidden layer was small, the difference between each training of the classifier was large. To make the results more representative, we took the average of ten runs, as shown in Fig. 24.

For a single-layer quantum classifier, there were 160 trainable parameters in the VQDNN. As the PQC line depth increased to 10 layers, the number of trainable parameters reached 430. Using the Adam optimizer, the learning rate was set to 0.1. It can be seen from the figure that as the depth of the quantum circuit increased,

the classifier can achieve better classification accuracy, and it can also converge faster. When the line depth was set to 10, the test accuracy can reach approximately 0.8. For a classic fully connected network with a single hidden layer, set the number of hidden layer nodes to 1, 2, 3, and 5, and the number of corresponding parameters to 1045, 2080, 3115, and 5185. Observing the experimental data, it can be seen that compared with the classical network with four-digit parameter number, the quantum classifier based on VQDNN still maintains significantly better classification performance under the number of trainable parameters that differ by an order of magnitude.

4.5. Comparison of VQDNN with other quantum neural network models

VQDNN experiments were conducted on the UCI and MNIST datasets, using three different procedures, i.e., VQDNN in low feature spaces, VQDNN in high feature spaces, and VQDNN in a

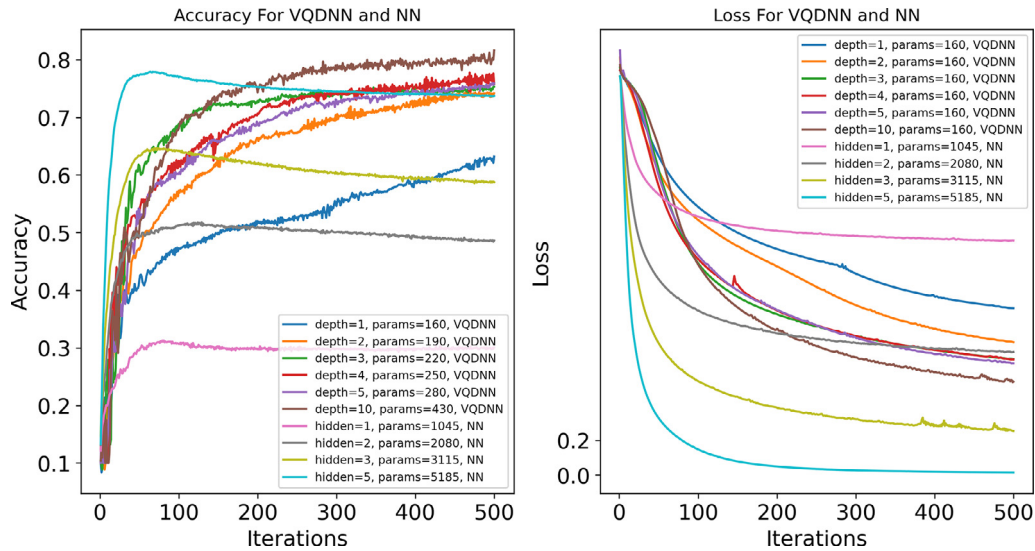


Fig. 24. VQDNN (left) and classical neural network (right) training loss curve. In these two figures, as the depth (the number of hidden layer nodes) increases, the faster the loss decreases.

sophisticated network structure. Finally, a comparative experiment was conducted on the UCI hand-written digit datasets for ten-label classification tasks. These experiments revealed that the proposed VQDNN had excellent classification performance and show that the VQDNN is superior to classical neural networks in some specific learning tasks under the limitation of the same number of trainable parameters. In this section, we first compared and analyzed the proposed model with other quantum neural network models, focusing on the performance of different quantum models on classification learning tasks. Then, theoretically analyzed the validity of the model.

Skolik et al. [43] proposed a quantum neural network model, using qubit encoding in combination with principal component analysis (PCA), which means that each image in MNIST is represented by a vector with the 10 components. They obtained an accuracy of approximately 0.73 in the binary classification task. Wei et al. [46] proposed a quantum convolutional neural network on NISQ devices. In their two-class ('0' and '8') and ten-class ('0' '9') image recognition experiments, the noise-free qCNN model obtained 0.963 and 0.743 classification accuracies, respectively. In Ref. [36] Potempa et al. achieved an accuracy of approximately 0.7. Johri et al. [18] designed a quantum nearest centroid classifier, and experimentally demonstrated it on an 11-qubit trapped-ion quantum machine. Finally, they achieved an accuracy of approximately 87.5% for the case of '2' vs. '7'. For the 10 different digits, they obtained an accuracy of approximately 77.5%. Lu et al. [28] downsampled '3' and '6' images into 8×8 images to fit the QCNN model with 6 qubits, and they obtained 0.9665 for the binary classification task.

Compared with the above five quantum neural network models, our experimental classification accuracy did have certain advantages, which depend on the different data encoding technologies, neural network structures, types of quantum gates, and model training technologies. Specifically, we believe that the following four practical implications were very important for the design of quantum neural network classifiers.

First, for high-dimensional data, in addition to using PCA, amplitude encoding can also be used to exponentially reduce the dimensionality, and then other data encoding methods can be selectively used according to the number of qubits available. Second, data encoding controls the expressiveness of quantum models. If the encoding is not "rich" enough, we may end up with

very limited model classes that can be expressed by variational circuits. Repeating encoding can help to increase the frequency spectrum, and thereby the expressivity of a quantum neural network model. Then, as additional layers are added to variational quantum circuits, the expressibility will increase under certain circumstances, but does not always continue to improve. The observation that different PQCs saturate at different layer numbers can inform the choice of PQC used in practice. Finally, to compare the classification performance (expressibility and entangling capability of the model), we concluded that if one is trying to design a PQC to increase expressibility, it is better to insert two-qubit gates such as R_z and R_y in pairs.

5. Conclusions

It is important to understand that, driven by the real dilemma of machine learning and the recent limitations of quantum computing, variational circuits have profoundly changed the research goals of quantum machine learning. They are not aimed at the acceleration of known models, but have produced a series of new models. Currently, their usefulness is still largely unknown and requires in-depth research. This shift in perspective has created many new research questions. Such as what type of model is a quantum circuit? How to choose an encoding method? Are their quantum properties such as superposition and entanglement useful? How can we best train them in practice? What is their generalization ability? This study will focus on some important preliminary answers to these questions.

We proposed a variational quantum deep neural network classification model and evaluate its classification performance using three VQDNN classifiers on two general datasets. The main findings of this study and the advantages of the model are as follows:

- Such a hybrid VQDNN architecture enabled us to build quantum machine learning algorithms capable of dealing with low-dimensional or high-dimensional inputs and potentially to implement these QML algorithms on NISQ devices with a limited number of qubits and shallow circuit depth.
- Numerical simulation results show that the proposed VQDNN models can achieve excellent classification accuracy and have better performance on the ten-label classification task than

their classical counterparts, which have the same tunable parameters, confirming that the VQDNN classifier can provide excellent classification ability.

- Additionally, we explored the way to build a more sophisticated classifier by stacking multiple PQC layers and mixing the encoding method together. This shows that stacking more layers in the VQDNN can obviously increase the model performance.
- The proposed VQDNN model can be applied in different quantum machine learning scenarios and potentially be implemented on NISQ devices, given the suitable encoding methods.

However, our research still has some shortcomings, as well as some work that can be carried out in the future.

- Extension of our model to more complicated datasets such as the Fashion-MNIST dataset and CIFAR-10 needs further in-depth research in the future to test its robustness and capability.
- Trying VQDNN with novel VQC architectures such as the quantum convolutional neural networks [31] to improve the performance of the current model is another further investigation.
- There is a fast growth in the number of qubits available in quantum computers [4], so it is necessary to implement this model on quantum hardware in the future.

CRediT authorship contribution statement

Yunqian Wang: Conceptualization, Methodology, Software, Writing – original draft. **Wei Huang:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

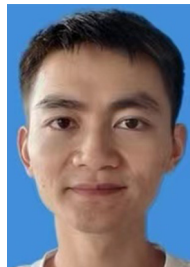
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